Determinants of Insurance Premiums: Lifestyle Choices vs. Demographic Factors

Varenya Katta, Raphael Karamagi, Tim Jun, Oat Twinvitoo

1. Introduction

1.1 Context and Research Question

In 2023, healthcare spending in the United States increased by 7.5%, totaling 4.9 trillion USD [8]. This increase in healthcare spending paralells increases in health insurance costs. As of recent health insurance costs have exceeded the growth rates of wages and inflation, making it increasingly difficult for individuals to afford coverage [7].

However, health insurance vary from individual to individual; factors such as age, location and tobacco use are legally allowed to increase the cost of one's premium. Further, individuals with other pre-existing health risks may also select costlier healthcare plans. Overall, we can categorize factors that potentially drive healthcare cost into two categories: Modifiable Lifestyle Choices (MLC) (behaviors an individual can change) and Baseline Demographic/Socioeconomic Factors (DRF) (fixed characteristics).

While previous studies have examined these variables individually, this study will attempt to bridge a gap by comparing the extent of variability attributable to each of these two categories: Does one category have a more substantial effect on insurance cost variance? The results may indicate how institutions and individuals can make healthcare more affordable. If insurance cost variance is heavily driven by MLC, customers can lower prices by modifying behaviors. However, if insurance cost is heavily driven by DRF, policy changes may be required to lower costs.

Dataset: We utilize the Medical Insurance Cost Prediction dataset from Kaggle [1]. The dataset contains approximately 100,000 observations including demographic details (Age, Sex, Income) and lifestyle indicators (Smoking status, BMI, Alcohol frequency). Constituent data was derived from three main sources: Centers for Disease Control and Prevention (CDC), U.S. Department of Health & Human Services (HHS), and Centers for Medicare & Medicaid Services (CMS). The response variable is the annual healthcare premium cost.

Modifiable Lifestyle Choices (MLC): Smoking status (Current, Former, Never), Alcohol consumption frequency, and Body Mass Index (BMI).

Baseline Demographics (DRF): Age, Sex, Education level, Income, Marital Status, and Household Size.

1.2 Exploratory Data Analysis

Table 1 provides summary statistics of the response variable (Premium), namely, median, mean, minimum and maximum. As the mean premium, at 582.32 USD, is substantially higher than the median, at 463.58 USD, we can infer a right-skewed distribution. Further, the extremely high value of the maximum premium, at 10962.55 USD, especially in comparison to the minimum premium value of 211.67 USD, may imply the presence of high outliers.

Table 1: Summary Statistics of Annual Premium Costs

Median Premium	Mean Premium	Min Premium	Max Premium
463.58	582.32	211.67	10962.55

2. Methodology

To determine whether the inclusion of lifestyle choices and demographics as predictor variables have an effect on the fit of a linear model, Hierarchical Multiple Linear Regression is employed [3]. This approach entails the comparison of a set of nested models: one full model will include the block of variables of interest while the other (reduced model) will include everything but said variables. Analysis of Variance (ANOVA) using Partial F-tests is used to determine if the addition of a block of variables significantly reduces the Sum of Squared Errors (SSE) compared to a reduced model.

We fit three distinct models:

Model 1: Baseline Demographics Accounts only for fixed factors the individual cannot easily change.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \cdots + \epsilon$$

Model 2: Lifestyle Only Accounts only for modifiable behaviors.

$$\ln(Y) = \beta_0 + \beta_{Smoker} + \beta_{Alcohol} + \beta_{BMI} + \epsilon$$

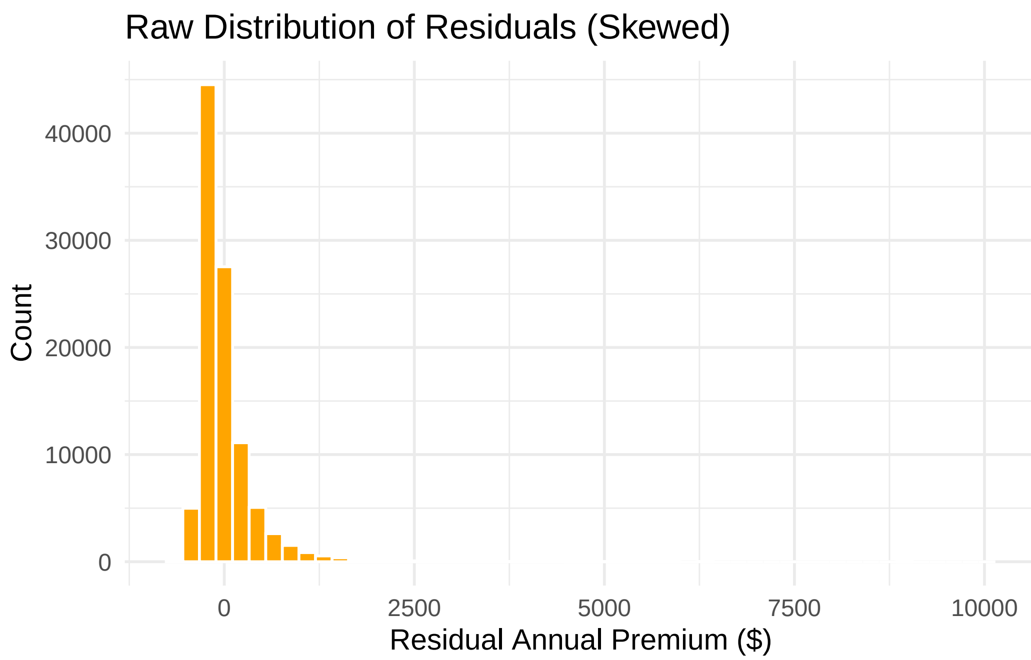
Model 3: Full Model (Demographics + Lifestyle) Combines both categories and includes the interaction term observed in EDA.

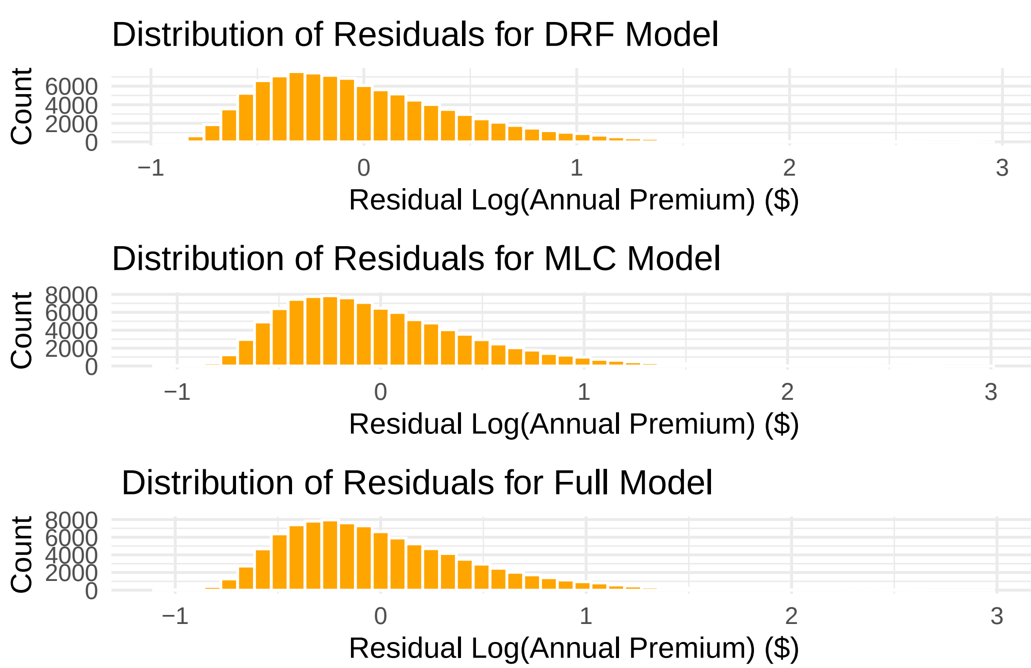
$$\ln(Y) = \text{Model } 1 + \text{Model } 2 + \epsilon$$

Following comparison using ANOVA, the change in R^2 between DRF vs. full model and the MLC vs. full model was utilized to assess the which set of variables had a more substantial effect in reducing the squared residuals.

2.1 Log Transformation

The use of log is to address the requirement of normality in the residuals necessitated by Least Squares Regression. When a full model is fit without log, its associated residual histogram is right skewed (Figure 1). To address this, a log-transformation [2] was applied to the outcome variable: annual premium. Doing so produces a more symmetric residual distribution (Figure 2, Bottom), and allows for the normality requirement to be satisfied. Other models (DRF and MLC) also showed roughly normal distributions when log was applied to the outcome variable.





2.2 Justification and Other Assumptions

We chose Ordinary Least Squares (OLS) regression on the log-transformed outcome. OLS provides highly interpretable coefficients, allowing for the quantification the changes in premiums associated with each lifestyle and demographic variable.

To ensure our results are valid, we will assess four key assumptions in the Results section:

1. Linearity: The relationship between predictors and the log-response is linear
2. Independence: We assume observations are independent (i.e., one individual's premium doesn't influence another's)
3. Equal Variance: The variance of residuals is constant across fitted values
4. Normality: The residuals are normally distributed

3. Results

3.1 Diagnostics and Assumptions Checking

To ensure the validity of our inference, we examined the residual plots for the full Model (Figure 3).

- Linearity & Equal Variance: The Residuals vs Fitted plot (Left) shows a random cloud of points around zero with no distinct pattern (e.g., no funnel shape). This indicates that the assumptions of linearity and constant variance are reasonably satisfied.
- Independence: One key source of this dataset is the National Health Interview survey; this survey is conducted on a voluntary basis. Consequently, there is a self-selection bias that may affect the ability to fully assume independence for this study.
- Normality of Residuals: As shown earlier in the residual histograms, the normality of the residuals for each of the three models is roughly fulfilled. Further, figure 4 (Right) shows the Normal Q-Q plot for the full model. The points (circles) adhere closely to the diagonal line for the majority of observations. However, the tails deviate upward. This "heavy-tailed" behavior indicates that our model slightly underestimates the premiums for the most extreme, high-cost outliers. Given the large sample size ($N \approx 100,000$), the Central Limit Theorem ensures that our coefficient estimates remain unbiased despite this deviation [2].

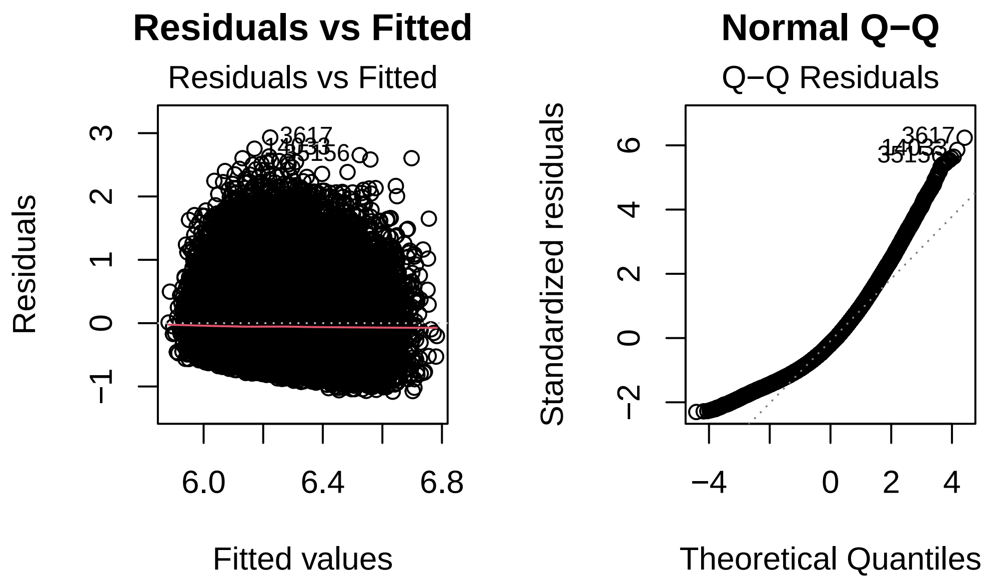


Figure 1: Diagnostic plots for the Full Model. Left: Residuals vs. Fitted values. Right: Normal Q-Q plot.

3.2 ANOVA Model Comparison

To test whether the addition of specific variable categories significantly improves the model, we utilized ANOVA F-tests.

Table 2: ANOVA Comparison: Demographics vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99980	22900.88	NA	NA	NA	NA
99972	22085.80	8	815.09	461.19	0

Table 3: ANOVA Comparison: Lifestyle vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99993	22683.72	NA	NA	NA	NA
99972	22085.80	21	597.92	128.88	0

Testing the Addition of Demographic Factors

H(null): *{Age}* = {Sex} = *{Region}* = {Income} = *{MaritalStatus}* = {Urban/Rural} = _{Education} = 0

H(alt): At least one of *{Age}*, {Sex}, *{Region}*, {Income}, *{MaritalStatus}*, {Urban/Rural}, _{Education} != 0

We compared the Lifestyle-only model against the Full model to see if adding Demographics significantly reduced the squared residuals. The F-test yielded an F-statistic of 461.19 and a p-value displaying 0, clearly less than <0.05. We reject the null hypothesis that all demographic coefficients are zero. Therefore, variation at least one of the socioeconomic/demographic factors seems to help explain the variation in the log(annual premium); the addition of DRF helps reduce the squared residuals and provide statistically significant explanatory power.

Testing the Addition of Modifiable Lifestyle Factors

H(null): $\beta_{Smoker} = \beta_{Alcohol} = \beta_{BMI} = 0$

H(alt): At least one of $\beta_{Smoker}, \beta_{Alcohol}, \beta_{BMI} \neq 0$

We compared the Demographic-only model against the Full model to see if adding Lifestyle factors improved the fit. The F-test yielded an F-statistic of 128.88 and a p-value displaying 0, clearly less than <0.05. We reject the null hypothesis that lifestyle coefficients are zero. Therefore, variation at least one of the modifiable life choices seems to help explain the variation in the log(annual premium); This confirms that modifiable choices (Smoking, BMI) also significantly explain variation in premiums.

The ANOVA tests conclude that the addition of MLC parameters to a DRF model may help explain the variation in the log(annual premium). However, as the degrees of freedom are not the same for each ANOVA test, the F value cannot be used directly to compare the impact of the added Demographic/Modifiable variables.

3.3 Comparison of R^2 Values

While ANOVA tests for statistical significance, comparing R^2 values allows us to assess the practical explanatory power (effect size).

Table 4: R^2 Values of Demographic, Lifestyle, and Full Model

Category	R2_Value
R^2 Demographic	0.0249303
R^2 Lifestyle	0.0341766
R^2 Full Model	0.0596348

Table 5: ΔR^2 of Demographic and Lifestyle Model Compared to Full Model

Category	R2_Value
ΔR^2 (Full Model - Demographic)	0.0347046
ΔR^2 (Full Model - Lifestyle)	0.0254582

The R^2 values are generally low across all models (0.0249 to 0.0596), indicating that the variables available in this dataset explain only a small fraction of the total variance in premiums.

However, we can compare the relative impact of the two categories. Adding Lifestyle variables to the demographic model increased the R^2 by roughly 0.0347. Conversely, adding Demographic variables to the lifestyle model increased the R^2 by only 0.0255. This suggests that modifiable lifestyle factors have a stronger marginal effect on predicting insurance costs than fixed demographic factors.

3.4 Regression Coefficients

Table 5 presents the coefficients for the Full Model. Since the response is log-transformed, coefficients (β) can be interpreted as an approximate $100 \times \beta\%$ change in the premium for small values.

Table 6: Coefficients for Modifiable Lifestyle Factors (Model 2)

Predictor	Estimate	Std. Error	P-Value
BMI	0.00562	0.0002976	<0.001
Smoker: Former	0.06373	0.0123031	<0.001
Smoker: Current	0.22166	0.0143493	<0.001
Alcohol: Occasional	-0.00062	0.0034994	0.859
Alcohol: Weekly	-0.00388	0.0042996	0.367
Alcohol: Daily	0.01245	0.0071756	0.083
age:Smoker: Former	0.00032	0.0002462	0.193
age:Smoker: Current	0.00087	0.0002870	0.002

Key Findings:

1. Smoking: Being a “Current Smoker” has a large positive coefficient (0.221). Holding other factors constant, current smokers pay approximately 22% more in premiums than never-smokers at the baseline age.

2. BMI: The coefficient for BMI is positive and statistically significant ($p < 0.001$). Higher body mass index is reliably associated with higher insurance costs.

3. Alcohol: The p-values for alcohol frequency are > 0.05. This suggests that, unlike smoking and BMI, self-reported alcohol frequency is not a statistically significant predictor of insurance premiums in this dataset.

4. Discussion

5. References

[1] Kaggle. (2024). Medical Insurance Cost Prediction Dataset. [2] Wooldridge, J. M. (2015). Introductory Econometrics. [3] UVa Library. (2023). Hierarchical Linear Regression. https://library.virginia.edu/data/articles/hierarchical-linear-regression [4] American Medical Association. (2024). National Health Expenditure Data. https://www.ama-assn.org/ [5] Bureau of Labor Statistics. (2023). Consumer Expenditure Survey: Health Insurance. [6] HealthCare.gov. (2024). How Insurance Companies Set Health Premiums. https://www.healthcare.gov/how-plans-set-your-premiums/ [7] Kaiser Family Foundation. (2023). Employer Health Benefits

Survey. https://www.kff.org/ [8] Centers for Medicare & Medicaid Services, Office of the Actuary. "National Health Expenditure (NHE) Fact Sheet: Historical NHE, 2023." U.S. Department of Health & Human Services